CAREERLY

AI-BASED RESUME ANALYZER AND PERSONALIZED INTERVIEW PRACTICE TRACKER

PROJECT SYNOPSIS

OF MINI PROJECT

BACHELOR OF TECHNOLOGY

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Proposed Topic: AI-Based Resume Analyzer and Personalized Interview Practice Tracker

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NOTATIONS / NOMENCLATURE

Symbol	Meaning
AI	**Artificial Intelligence**
API	**Application Programming Interface**
AST	**Abstract Syntax Tree**
CWE	**Common Weakness Enumeration**
LLM	**Large Language Model**
UI	**User Interface**
URL	**Uniform Resource Locator**

1. INTRODUCTION

1.1 Project overview

Applying for a job involves more than preparing a resume. Candidates must understand the requirements of a role, present relevant experience clearly, and communicate their knowledge in an interview. These tasks can be difficult to coordinate, especially for students and early-career job seekers who may have limited access to individualized career guidance.

The proposed project, AI-Based Resume Analyzer and Personalized Interview Practice Tracker, is a web application intended to connect resume preparation with interview practice. A user provides a resume and a target job role or job description. The planned complete system will extract relevant information from the resume, compare it with the role requirements, and present matching strengths and possible skill or evidence gaps. Based on the selected role and analysis, the system will provide relevant interview questions. Users will submit text answers, receive structured feedback, and review their practice history and areas for improvement in a dashboard.

The project aims to make preparation more focused: resume insights should inform what a candidate practises, and practice results should help the candidate decide what to work on next. Scores and feedback are intended as preparation aids, not as guarantees of recruitment outcomes or substitutes for professional hiring decisions.

1.2 Field and technologies

The project is in the field of web application development for career readiness, with planned use of Artificial Intelligence and Natural Language Processing for resume-to-role comparison and interview feedback. The current prototype is implemented using HTML5 for page structure, CSS3 for styling and responsive layouts, and vanilla JavaScript for front-end interactions. The prototype contains an overview dashboard, resume analyzer form, interview practice screen, and progress view.

The proposed full system will require server-side components for resume text extraction, AI service calls, authentication, and persistent storage. Those components have not yet been implemented. The current prototype does not send files to a server, parse resume content, call an AI model, or save user sessions. The analysis scores and answer feedback shown in the interface are illustrative examples only.

1.3 Technical terms

Resume parsing is the extraction of structured information, such as skills, education, and work experience, from an uploaded document. Role matching compares extracted resume information with the requirements in a target job description. Natural Language Processing refers to computational processing of human language and may be used in a future implementation to support this

comparison and answer feedback. An Applicant Tracking System is software employers may use to manage recruitment and candidate information. The term readiness score in this project refers to a planned summary of practice performance; it is not an objective prediction of whether a candidate will be hired.

2. LITERATURE REVIEW

Existing career-preparation services address parts of the proposed workflow. Jobscan describes a resume scanner that compares a resume with a job description and reports a match rate using criteria including hard skills, education, job title, soft skills, and keywords [1]. This demonstrates the value of comparing application materials with a specific vacancy rather than assessing a resume in isolation. The proposed project follows this role-specific approach, while intending to carry the identified strengths and gaps forward into interview practice.

Google's career guidance provides interview preparation advice, including guidance on preparing a clear, concise response to common interview questions [2]. This illustrates the importance of structured preparation and communicating relevant experience. The proposed application adds an interactive practice flow and a place to review session progress alongside resume preparation.

These resources provide useful examples of resume tailoring and interview preparation, but the current prototype is not a replication of their services and does not claim to match their capabilities. The projectâ€™s intended contribution is to bring resume-to-role analysis, role-related text interview practice, feedback, and progress tracking together in one workflow. The current prototype demonstrates the screens and navigation for this workflow; it has not yet implemented the underlying AI, resume parsing, or persistent tracking.

Because resumes contain personal information and are uploaded as files, secure handling is also an important part of the planned system. OWASP recommends layered file-upload controls, including allow-listing necessary file types, validating actual file content rather than trusting the supplied MIME type, limiting file size, and storing uploads with appropriate access restrictions [3]. The current browser-side extension and size checks are only preliminary user-interface checks; they are not sufficient security controls for a deployed application.

The front-end prototype uses established browser technologies. MDN Web Docs describes HTML as the language for structuring web content, CSS as the language for styling it, and JavaScript as a programming language used to add behavior to web pages [4]â€“[6]. These technologies were selected for the prototype because they support a lightweight interface without a framework. A future production version will also need a server, database, secure file storage, and an AI service integration.

3. OBJECTIVES

The general objective is to develop a web-based application that helps students and job seekers connect resume improvement with personalized interview preparation and progress tracking.

The specific objectives are:

- Design a responsive interface for an overview dashboard, resume analysis, interview practice, and progress history.
- Allow a user to select a resume and provide a target job title and optional job description.
- In the planned complete system, extract relevant resume information and compare it with role requirements.
- Present relevant strengths and skill or evidence gaps in a clear and understandable form.
- Provide interview questions related to a user's target role and identified practice areas.
- Accept text-based interview answers and provide structured feedback using a consistent evaluation rubric.
- Track practice sessions and summarize performance and focus areas over time.
- Apply suitable privacy and file-handling safeguards when server-side upload and analysis are implemented.
- Validate the principal interface flows in the current HTML, CSS, and JavaScript prototype.

4. METHODOLOGY / PLANNING OF WORK

The project will be developed incrementally in the following stages:

1. Requirement and workflow definition: Identify user needs, define the candidate journey, and document the functional and non-functional requirements.

2. Interface design: Plan the dashboard, resume analysis, interview practice, and progress screens, including responsive behavior.

3. Front-end prototype: Implement the screens and navigation in HTML and CSS, and add client-side interactions in JavaScript. The existing prototype demonstrates file selection and basic extension/size checks, sample analysis, a three-question practice flow, demo feedback, and progress screens.

4. Backend and data design (planned): Select an API, database, authentication, and private file-storage approach. Design data entities for users, resumes, target roles, practice sessions, answers, and feedback.

5. Resume processing and AI integration (planned): Add server-side file validation and resume text extraction; then integrate an AI service to compare resume content with job requirements, generate relevant questions, and assess answers using a defined rubric.

6. Testing and refinement: Test validation, navigation, responsive layouts, errors, data access, and end-to-end flows. Review the clarity and usefulness of feedback and improve the interface based on testing.

7. Documentation and demonstration: Record the implemented features, limitations, setup instructions, and demonstration workflow.

The first prototype stage is implemented. Backend, database, secure upload, resume parsing, and AI integration remain planned work. Resume processing should be performed server-side with file type and content validation, size limits, restricted access, and secure storage in line with applicable upload-security guidance [3].

5. REFERENCES

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